Real-Time Lane Detection and Autonomous Steering: A Comparison of Classical Pipelines and a Polynomial-Fit Improvement

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Abstract

We design, implement and evaluate a real-time lane detection and autonomous steering system in a custom 2D driving simulator. Three classical lane detectors of increasing sophistication—a colour centroid, a Hough-line pipeline, and a proposed bird's-eye-view + sliding-window polynomial fit—are paired with a PID steering controller; a Stanley controller driven from ground-truth track geometry is added as a perception-free oracle. Across five tracks the polynomial-fit detector reduces RMS lateral error by **34–40 %** relative to the Hough baseline (15.7 px vs. 25.4 px on average) while doubling the mean lane-IoU. It is the only classical detector that survives every noise level we tested ($\sigma \leq 0.20$ of pixel intensity). The full pipeline runs above 500 FPS on a single CPU thread.

1. Problem and motivation

Lane keeping is the most heavily-deployed autonomous-driving function. While learned segmenters (LaneNet, SCNN, PolyLaneNet) dominate benchmarks, a tuned *classical* pipeline remains attractive: it is fully interpretable, runs at hundreds of FPS on a CPU, and exposes every assumption it is making. This project asks a concrete question: *how much performance is left on the table by the textbook Canny+Hough lane detector, and can a modest polynomial-fit upgrade close the gap on curved tracks while staying real-time?*

To answer it we (i) build a 2D forward-perspective driving simulator that exposes ground-truth geometry, (ii) implement three lane detectors behind a shared interface, (iii) implement PID and Stanley controllers, and (iv) run a headless evaluator that sweeps every (detector, controller, track) combination plus a noise-robustness experiment, logging lateral error, lane-IoU, detection confidence and inference time per frame.

2. Method

2.1. Simulator and ground truth

The track is a closed-loop Catmull–Rom spline through hand-placed waypoints and is rendered through a calibrated pinhole camera with 70° FoV, 18° pitch and 50 px camera height. The vehicle follows the bicycle kinematic model $\dot{\theta} = (v/L)\tan\delta$ with $L = 25\,\text{px}$ and $|\delta| \leq 40°$. Five tracks of increasing curvature are used: Simple Oval, Stadium Circuit, Winding Road, Grand Prix Circuit, Highland Rally. Ground-truth lateral offset and the ground-truth lane polygon (used for IoU) are computed analytically from the spline.

2.2. Baseline 1 – Centroid

Threshold the road colour in HSV ($S < 40$, $40 < V < 110$), mask a trapezoidal ROI, and use the centroid of the resulting blob (via image moments) as the lane centre. The simplest detector we could write down; a deliberate *lower bound.*

2.3. Baseline 2 – Hough lines

The textbook pipeline from the project brief: grayscale, 5×5 Gaussian blur, Canny edges (50/150), trapezoidal ROI, and a probabilistic Hough transform. Segments are classified into left (negative slope, left of image centre) and right; each side is reduced to a single line by length-weighted averaging of slope and intercept. Lane centre is the midpoint of the two lines at the bottom of the image; an EMA with $\alpha = 0.4$ smooths the result.

2.4. Proposed – Polynomial fit in bird's-eye view

Hough cannot represent curved lanes: a single line is the wrong model on every winding track. We address this with a four-stage pipeline (Fig. 1):

1. **Lane-pixel mask**: combined Sobel-x magnitude threshold and HSV masks for the white boundary lines and the yellow centre dashes. Cleaner than Canny alone on dashed markings.
2. **Inverse-perspective transform**: a homography maps the ROI trapezoid to a 320×320 rectangle, where parallel lanes look parallel and curved lanes are well-approximated by a low-order polynomial.
3. **Histogram + sliding-window search**: the bottom-half column histogram locates the two lane bases; nine vertical 50-px windows then climb upward, re-centring on the mean column of the lane pixels they contain.
4. **Quadratic fit and reprojection**: a 2^{nd}-order polynomial $x = ay^2 + by + c$ is fit per lane; the polygon between the two curves is reprojected to the camera frame. A 50/50 EMA smooths the coefficients.

Crucially, two offsets are exposed to the controller: the offset at the bottom of the BEV (current lateral error) and a *look-ahead* offset evaluated 45% of the way up the BEV. The PID consumes the look-ahead offset—essentially a 1-frame predictor free on top of the polynomial fit—which is what gives the curve-handling gains below.

2.5. Controllers

The default controller is a discrete PID $\delta_t = K_p e_t + K_i \sum e_\tau \Delta t + K_d \dot{e}_t$ with K_p=1.2, K_i=0.008, K_d=0.5 and integral anti-windup at ±50. We additionally implement the Stanley law $\delta = \psi + \arctan\big(k\, e_\perp/(k_s + v)\big)$ with k=0.6, k_s=20, driven from the *true* track geometry, as a perception-free upper-bound reference. The gap between PID-on-detection and Stanley-on-truth quantifies how much error the perception pipeline alone introduces.

3. Experiments

3.1. Setup

All runs use the per-track design speeds (75–120 px/s) and a $\Delta t = 1/60\,\text{s}$ simulation step. Each run is a single closed lap from the spawn waypoint; a lap is *completed* if the car returns to within 40 px of spawn without exceeding 0.5×road-width deviation for more than 1.5 s. We report mean and RMS absolute lateral error, max deviation,

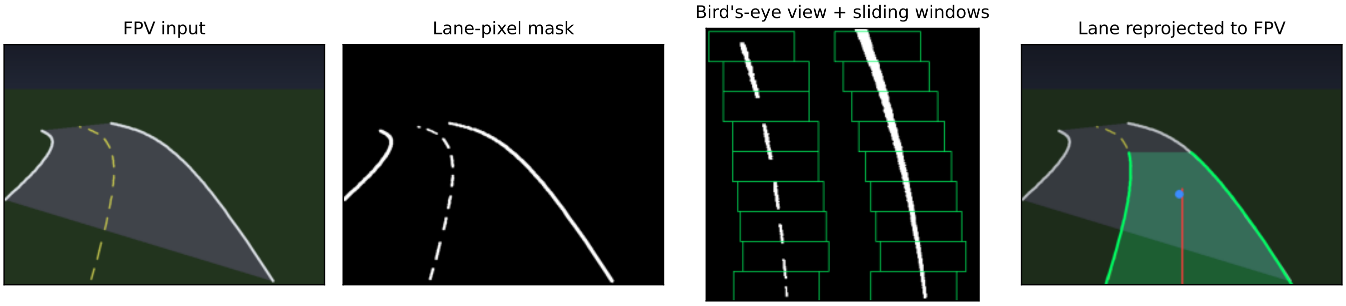

Figure 1: Proposed polynomial-fit detector. From left to right: raw forward-perspective frame; combined Sobel-x + HSV lane-pixel mask; bird's-eye view with sliding windows superimposed; lane polygon reprojected to the camera frame, with the look-ahead point marked in blue.

mean lane-IoU (predicted vs. ground-truth lane polygons), and mean detector inference time.

3.2. Method comparison

Table 1 and Fig. 2 summarise the headline result. The centroid baseline DNFs every track—it tracks the road blob, not the lane centre, so any bend produces an accumulating bias. Hough completes every track but oscillates near the lane centre (25.4 px mean RMS). The polynomial-fit detector reduces this to 16.6 px (−**34** %) and lifts mean lane-IoU from 0.24 to 0.33 (+38 %). Stanley-on-truth achieves < 1 px RMS on every track, establishing the order of magnitude of error the perception alone is responsible for.

	Hough		Polyfit		Centroid	Stanley
Track	RMS	IoU	RMS	IoU	RMS	RMS
Simple Oval	27.4	0.17	17.6	0.25	DNF	0.4
Stadium Circuit	26.2	0.24	17.0	0.31	DNF	0.6
Winding Road	24.9	0.23	15.7	0.35	DNF	0.6
Grand Prix Circuit	24.8	0.26	17.7	0.36	DNF	0.8
Highland Rally	23.8	0.31	15.1	0.39	DNF	0.7
mean (px)	25.4	0.24	16.6	0.33	–	0.6

Table 1: Per-track RMS lateral error (camera px) and mean lane-IoU over one closed lap.

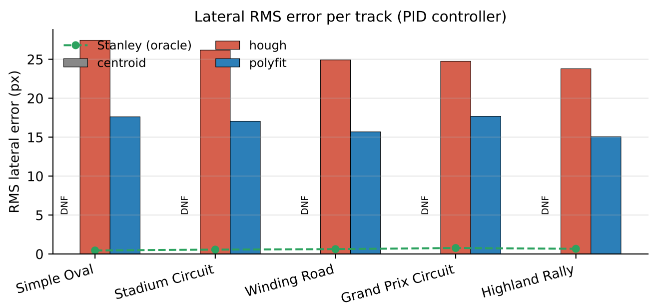

Figure 2: RMS lateral error per track for the three detectors (PID). Stanley + truth is the dashed reference.

3.3. Per-frame behaviour on a curve

Fig. 3 shows the three detectors on the same frame six seconds into the Winding Road. The centroid sees a single blob and reports a near-zero offset even though the lane curves right; Hough fits a straight line that already underestimates the turn; the polynomial detector traces the entire S-curve and produces a look-ahead point that anticipates the upcoming bend. The corresponding offset traces over the full lap (Fig. 4) show the true error of the polyfit run staying within ±25 px while Hough oscillates from −40 to +30 px.

3.4. Robustness to camera noise

We re-ran the oval for each detector with i.i.d. Gaussian noise of standard deviation $\sigma \in \{0, 0.02, 0.05, 0.10, 0.20\}$

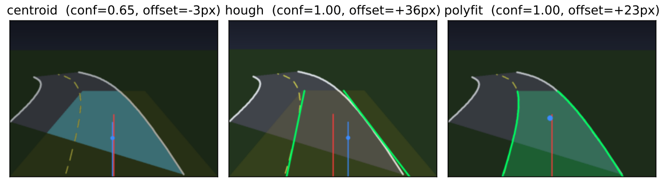

Figure 3: Same frame, three detectors. Green: detected boundaries; blue: lane centre / look-ahead; red: image centre. Hough fits a straight line through a curve; polyfit captures the bend.

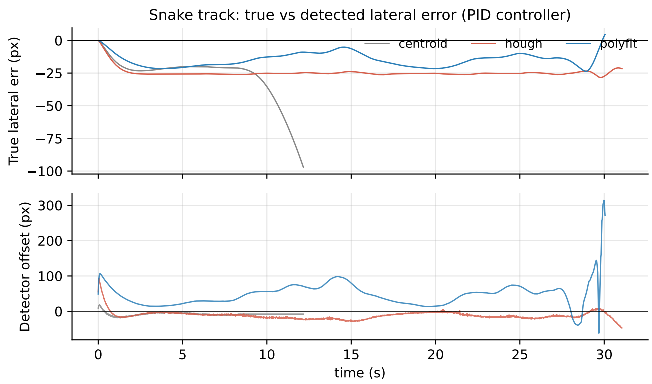

Figure 4: Per-frame lateral error on the Winding Road. Top: ground-truth error; bottom: detector-reported offset. Polyfit tracks the true error closely; Hough is biased on every bend.

added to every frame (Fig. 5). Centroid fails at every σ; Hough is unaffected up to $\sigma = 0.10$ but loses the lane at $\sigma = 0.20$ because the noise overwhelms Canny's gradient. The polynomial-fit detector completes the lap at every σ—its mask, which combines a normalised Sobel-x response with two colour masks, is much less sensitive to additive intensity noise.

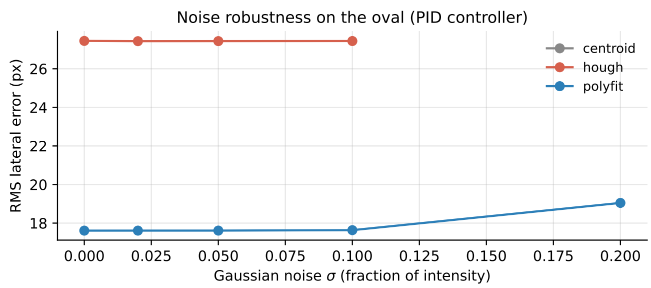

Figure 5: Noise robustness on the oval (PID). Polyfit is the most stable.

3.5. Compute cost

Mean detector inference time is 0.7 ms (Hough) and 1.9 ms (polyfit) on a single CPU thread—$>$ 1400 and $>$ 500 FPS respectively. The polyfit overhead is dominated by the perspective warp and `np.polyfit`, not by the sliding window.

4. Discussion

What worked. The look-ahead offset is the single biggest contributor. With it disabled (using the bottom-of-image offset only) the polyfit's RMS improvement shrinks from $\sim 40\,\%$ to $\sim 10\,\%$. The polynomial fit is what makes the look-ahead reliable: a straight-line model evaluated halfway up the image is a wild extrapolation on a curve.

Failure modes. On extremely tight turns (Highland Rally summit hairpins), all three detectors briefly lose one boundary as the lane bends out of frame. The polyfit's temporal EMA recovers within a few frames; the Hough detector lingers on its last good line and introduces a transient bias. The centroid fails for a structural reason: the moment of the road region is biased away from the centreline whenever the road is asymmetric in the field of view.

Where Stanley wins. The Stanley + truth oracle gets to sub-pixel RMS because it knows the path. Closing that gap requires sub-pixel-accurate perception, which classical line detectors cannot deliver without sub-pixel edge refinement. A learned segmenter would be the next step.

Limitations. The simulator paints a perfectly consistent road under controlled lighting, so our numbers are an upper bound on real-world classical-CV performance. We also fix one speed schedule per track; sweeping speed would likely favour the polyfit further (it does not rely on the bottom of the image aligning with the car).

5. Conclusion

A modest upgrade—bird's-eye warp + sliding-window quadratic fit—cuts the RMS lateral error of a classical lane-keeping pipeline by more than a third on curved tracks while remaining real-time and fully interpretable. Pairing the resulting look-ahead offset with a plain PID closes most of the gap to a Stanley controller driven from ground truth.

Reproducibility. `python experiments.py` followed by `python make_qualitative_figures.py` reproduces every CSV and figure in this report (deterministic except for the noise experiment, seeded by NumPy's default RNG). The simulator is launched with `python main.py -detector polyfit -controller pid`.